

Sparse convex hulls for network flows coupled to a simplex

Abstract

We study the convex hull of selected bilinear products between a bounded equality-constrained network flow and a simplex vector. The network and the retained product coordinates provide two distinct sources of structure: circulations factor across undirected blocks, and each block need retain only the simplex states observed there. We develop exact formulations and original-coordinate descriptions from these reductions, and examine the limits of small-coefficient descriptions. The starting point is classical simplex-vertex disaggregation; the structural question is how much of that formulation is needed for a sparse set of products.

1 Introduction

Network models often combine continuous routing decisions with a choice among mutually exclusive services or operating modes. A route-specific adjustment to the cost of a service gives a product of an arc flow and a service weight. Only a small fraction of the possible route-service pairs may occur in the model. Convexifying each product independently discards restrictions linking flows through the network and linking service weights through the simplex. An exact joint hull retains these restrictions, but its standard extended formulation introduces a copy of every arc flow for every simplex vertex.

This paper studies the structure of that joint hull when only selected products are retained. We distinguish the size of an extended formulation from the coefficients and separation cost of an original-coordinate description. We also distinguish equality-constrained flow models from network models with balance inequalities: adding slack arcs to convert the latter to the former can change the graph structure on which a specialized result depends. The elementary foundations below make the state and graph decompositions explicit, including degenerate capacities and zero simplex weights.

2 Model, prior work, and disaggregation

2.1 The sparse network-simplex hull

Let $D = (V, E)$ be a finite directed multigraph. Parallel arcs and self-loops are permitted. Its incidence matrix $A \in \{-1, 0, 1\}^{V \times E}$ has a -1 at the tail and a $+1$ at the head of a non-loop arc; a self-loop has a zero column. Thus $Ax = b$ uses the convention *incoming minus outgoing*. For $b \in \mathbb{Q}^V$ and finite $u \in \mathbb{Q}_+^E$, set

$$P = \{x \in \mathbb{R}^E : Ax = b, 0 \leq x \leq u\}. \quad (1)$$

Lower bounds can be handled by translating x and the corresponding product coordinates. Throughout, graph assumptions concern the graph in (1) after any such modeling transformations.

For an integer $m \geq 0$, write $[m] = \{1, \dots, m\}$ and

$$\Delta_m = \{y \in \mathbb{R}_+^m : \mathbf{1}^\top y \leq 1\}, \quad \lambda_j = y_j \ (j \in [m]), \quad \lambda_0 = 1 - \mathbf{1}^\top y. \quad (2)$$

The *states* are the $d = m + 1$ vertices $e^0 = 0, e^1, \dots, e^m$ of Δ_m ; e^j is the j th unit vector when $j \geq 1$. The residual state is state 0. An observation set $\mathcal{O} \subseteq E \times [m]$ specifies the products retained as original coordinates. We study

$$S = \{(x, y, z) : x \in P, y \in \Delta_m, z_{ej} = x_e y_j \quad ((e, j) \in \mathcal{O})\}, \quad (3)$$

$$H = \text{conv } S. \quad (4)$$

An unobserved product is absent from z , even if it is introduced temporarily in an extended formulation. State 0 has no observed coordinates. We use $\text{conv } \emptyset = \emptyset$. When $m = 0$, y and z are empty vectors and $H = P$.

The original coordinates are (x, y, z) . An extended formulation is a linear system in these coordinates and additional continuous variables whose projection is H . A separation procedure must either certify membership or produce a linear inequality valid for all of H and strictly violated by the candidate. Unless specified otherwise, a facet is taken relative to the affine hull. A bound on integer coefficients presupposes a stated normalization; a ratio of two nonzero coefficients is invariant under scaling the inequality. Arithmetic-operation bounds are distinguished from bounds on rational encoding length.

For comparison, the usual independent product relaxation adds, for every $(e, j) \in \mathcal{O}$, the four inequalities

$$0 \leq z_{ej}, \quad z_{ej} \leq u_e y_j, \quad z_{ej} \leq x_e, \quad z_{ej} \geq x_e - u_e(1 - y_j) \quad (5)$$

to $x \in P$ and $y \in \Delta_m$. These are the McCormick inequalities for the box $[0, u_e] \times [0, 1]$ [11]. They need not describe the joint hull in (4).

2.2 Relation to prior work

The general convexification principle is established. Disjunctive hull formulations [3] apply after expanding the simplex into its vertices. Simultaneous convexification over a polytope times a simplex is developed by Davarnia et al. [5]. Khademnia and Davarnia [8] specialize this setting to network polytopes: their electronic companion, p. 1, equation (25), gives a full extended hull,¹ and their aggregation-and-projection framework gives a complete original-space description in principle. Their explicit tree and forest constructions identify more structured inequality families; the forest recipe for multiple simplex coordinates should not be confused with the entire projection framework. Their general framework permits balance inequalities, and their Section 3 represents equality balances by pairs of opposite inequalities. Thus their network specialization includes (1); here we exploit its underlying equality circulation space. We use the published article and its electronic companion for equation and example numbering.

The shared-simplex gluing principle is also known. In particular, Davarnia [4, Proposition 2.6] proves that convexifying Cartesian components separately is exact when the components share the same simplex. The cited locator belongs to the dissertation, which is distinct from the 2017 journal article. Our block decomposition applies this principle to the independent circulation spaces of an equality-constrained network. We give a direct proof to make the treatment of sparse observations and zero weights explicit.

Several established methods explain the subsequent structural reductions. Reduced reformulation–linearization uses multiplied linear equations to remove common-factor products [10]. In networks, the remaining freedom can be identified with circulations supported on unobserved arcs. The network representation of Cayley hulls of Kis and Horváth [9] gives cut descriptions and transportation

¹The published electronic companion is available at [this direct supplement link](#).

projections; their Section 5.7 is a close predecessor for the parallel-path construction. The support geometry of Minkowski sums and edge-generated zonotope refinements is classical [7, 12]. These are ingredients of the graph-specific formulations and separation procedures, rather than new general convexification mechanisms.

Two further comparisons delimit coefficient obstructions. Three-way transportation universality is established by De Loera and Onn [6], and large coefficients in general 0/1 polytopes are classical [2]. A restricted network realization must therefore be specified as part of a coefficient result. Moreover, the integrality of *aggregate* flows on two-terminal series-parallel networks does not give a hull description in selected state-flow coordinates. Almoghrabi et al. [1, Remark 1] explicitly distinguish aggregate integrality from decomposability of the vector of individual commodity flows. Our observed products retain such individual-state information. Neither a sparse extended formulation nor polynomial separation precludes large numerical coefficients after projection.

2.3 The known exact disaggregation

For $\lambda \geq 0$, define the scaled flow system

$$P(\lambda) = \{f \in \mathbb{R}^E : Af = \lambda b, 0 \leq f \leq \lambda u\}. \quad (6)$$

For $\lambda > 0$, this is λP ; at $\lambda = 0$ it is $\{0\}$, including when P is empty. Using (6) avoids assigning an ambiguous meaning to $0\emptyset$.

Proposition 2.1 (Simplex-vertex disaggregation). *A point (x, y, z) belongs to H if and only if $y \in \Delta_m$ and there exist state flows $f^0, \dots, f^m \in \mathbb{R}^E$ such that*

$$f^k \in P(\lambda_k) \quad (k \in \{0, \dots, m\}), \quad \sum_{k=0}^m f^k = x, \quad f_e^j = z_{ej} \quad ((e, j) \in \mathcal{O}). \quad (7)$$

The equivalence holds even if P is empty. If the system is feasible, it provides a convex decomposition into at most $m + 1$ points of S .

Proof. For a point of S , set $f^k = \lambda_k x$. This satisfies all displayed relations, since $\sum_k \lambda_k = 1$. The system is linear in (x, y, z, f) , so its projection is convex and contains H .

Conversely, suppose the system is feasible. If $\lambda_k = 0$, the finite upper bounds give $f^k = 0$. For each positive weight let $x^k = f^k / \lambda_k \in P$ and define the graph point $p^k = (x^k, e^k, z^k)$ by $z_{ej}^k = x_e^k(e^k)_j$. Its weighted average has x coordinate $\sum_k f^k = x$, y coordinate $\sum_k \lambda_k e^k = y$, and observed product coordinate $\lambda_j x_e^j = f_e^j = z_{ej}$ when $\lambda_j > 0$. If $\lambda_j = 0$, that observed coordinate is zero on both sides. Thus $(x, y, z) = \sum_{k: \lambda_k > 0} \lambda_k p^k \in H$. There is at least one positive weight, so feasibility also implies $P \neq \emptyset$. This proves the empty-set assertion. $\square$

The formulation has $(m + 1)|E|$ state-flow variables before elimination. Its rational size is polynomial in the ordinary network and simplex input, so generic polynomial-time hull optimization and separation already follow from linear programming. The structural issue is avoiding this full state-by-arc expansion, or projecting it into useful original-coordinate inequalities. In particular, eliminating $f^0 = x - \sum_{j=1}^m f^j$ keeps the residual bounds

$$0 \leq x - \sum_{j=1}^m f^j \leq \left(1 - \sum_{j=1}^m y_j\right)u. \quad (8)$$

These bounds cannot in general be discarded.

3 Independent blocks and locally merged states

3.1 Circulation factorization

All graph connectivity terminology below refers to the undirected multigraph underlying D . Isolated vertices impose only their balance equation. Let E_{br} be the bridges. Let $\mathcal{B}$ be the cyclic blocks: the maximal biconnected edge blocks that contain a cycle, with each self-loop treated as its own block. A pair of parallel edges can form a cycle of length two. The sets E_{br} and E_B for $B \in \mathcal{B}$ partition E . The cycle rank of a connected block is

$$r_B = |E_B| - |V_B| + 1. \quad (9)$$

In particular, a loop has rank one. Rank refers to the ambient circulation space, even when capacities reduce the dimension of the feasible polytope.

The equation $Av = b$ is solvable exactly when the entries of b sum to zero on every connected component. Necessity follows by summing the incidence rows in a component. For sufficiency, choose a spanning forest, set the nonforest coordinates to zero, and determine each tree-edge flow by the balance sum on one side of its fundamental cut, with the sign determined by the arc orientation. This satisfies every balance equation. This construction uses $O(|V| + |E|)$ arithmetic operations, and for rational b produces a rational v of polynomial encoding length. The reference flow v need not obey its capacity bounds. If the component condition fails, P and H are empty.

Fix such a reference vector v . For each B , let A_B be the incidence matrix restricted to columns in E_B and rows in V_B , and define

$$K_B = \{h \in \mathbb{R}^{E_B} : A_B h = 0, -v_{E_B} \leq h \leq u_{E_B} - v_{E_B}\}. \quad (10)$$

Unlike P , this is a domain of circulation *deviations*; it need not contain the origin.

Lemma 3.1 (Block factorization). *Every $g \in \ker A$ vanishes on E_{br} and restricts to a circulation on each cyclic block. Conversely, combining arbitrary block circulations and zero bridge coordinates gives an element of $\ker A$. Consequently, if every bridge satisfies $0 \leq v_e \leq u_e$, the map*

$$x_e = v_e \quad (e \in E_{\text{br}}), \quad x_{E_B} = v_{E_B} + h_B \quad (B \in \mathcal{B}) \quad (11)$$

is an affine bijection from $\prod_{B \in \mathcal{B}} K_B$ to P . If a bridge bound fails, P is empty. The product of an empty family of block domains is a singleton.

Proof. Choose a spanning forest of the underlying multigraph, omitting loops. Each nonforest edge gives a signed fundamental-cycle circulation: use the unique forest path between its ends, or just the edge itself for a loop. These vectors form a basis of $\ker A$. Indeed, subtracting from a circulation the combination that cancels all nonforest coordinates leaves a circulation supported on a forest, which is zero by successive leaf elimination.

Each fundamental cycle lies in a single cyclic block. Hence its vector is zero on every bridge, and on every block it is either zero or a circulation of that block. The basis expansion proves the first assertion. For the converse, extend each block circulation by zero to E . Its incidence is zero at every vertex, including articulation vertices, so their sum is a global circulation. The sum is direct because the block edge sets are disjoint. Applying this decomposition to $x - v$ proves that $Ax = b$ is equivalent to the circulation conditions and the fixed bridge coordinates in (11). The bounds then separate by edge block, giving precisely (10) and the bridge conditions. $\square$

In particular, an articulation vertex does not carry an additional coupling variable between block deviations. Its fixed net balance is already carried by v ; every block deviation has zero net balance at that vertex separately. Any spanning tree in B gives a fundamental-cycle matrix $C_B \in \{0, \pm 1\}^{E_B \times r_B}$ with chord rows equal to the identity. Thus $h_B = C_B t_B$ gives unique cycle coordinates $t_B \in \mathbb{R}^{r_B}$. These coordinates remain useful when K_B is lower-dimensional.

3.2 The state merger and its exact refinement

For a nonempty convex set K , use the usual scalar multiplication $\alpha K = \{\alpha h : h \in K\}$, so $0K = \{0\}$. For nonnegative scalars,

$$\alpha_1 K + \cdots + \alpha_q K = (\alpha_1 + \cdots + \alpha_q) K. \quad (12)$$

If the total weight is positive, the inclusion from left to right follows by dividing by that weight and taking a convex combination; the reverse inclusion uses the same point of K in each summand. If the total is zero, both sides are $\{0\}$. An empty sum is also interpreted as $\{0\}$.

For each block define its observed labels and their count by

$$J_B = \{j \in [m] : (e, j) \in \mathcal{O} \text{ for some } e \in E_B\}, \quad a_B = |J_B|. \quad (13)$$

This is a structural definition: a label in J_B remains present when $y_j = 0$. Let $*_B$ denote a local merged state, distinct from the global residual state 0, and set

$$I_B = J_B \cup \{*_B\}, \quad \lambda_{B,j} = y_j \ (j \in J_B), \quad \lambda_{B,*} = 1 - \sum_{j \in J_B} y_j. \quad (14)$$

The merged state collects all labels absent from J_B and state 0. Different blocks may merge different global states.

Theorem 3.2 (Exact blockwise state reduction). *Assume $P \neq \emptyset$. A point (x, y, z) belongs to H if and only if $x \in P$, $y \in \Delta_m$, every observed bridge satisfies*

$$z_{ej} = v_e y_j \quad (e \in E_{\text{br}}, (e, j) \in \mathcal{O}), \quad (15)$$

and for every cyclic block B there exist vectors $h^{B,k} \in \mathbb{R}^{E_B}$, $k \in I_B$, satisfying

$$h^{B,k} \in \lambda_{B,k} K_B \quad (k \in I_B), \quad (16)$$

$$\sum_{k \in I_B} h^{B,k} = x_{E_B} - v_{E_B}, \quad (17)$$

$$h_e^{B,j} = z_{ej} - y_j v_e \quad ((e, j) \in \mathcal{O}, e \in E_B). \quad (18)$$

The block conditions yield state flows in (7), and hence a decomposition into at most $m + 1$ graph points, by explicit proportional refinement.

Proof. By lemma 3.1, every K_B is nonempty. Start with state flows in (7). Their deviations $g^k = f^k - \lambda_k v$ are circulations. On B , they satisfy $g_{E_B}^k \in \lambda_k K_B$: division proves this for positive weights, and both deviations and scaled domains are zero for zero weights. For $j \in J_B$ take $h^{B,j} = g_{E_B}^j$, and sum the other deviations into $h^{B,*}$. Identity (12) proves (16). Summing the deviations proves (17), and the observed coordinates give (18). On bridges the deviations vanish, proving (15).

Conversely, fix feasible local vectors. For each global state k and block B , if $k \in J_B$, set $g_{E_B}^k = h^{B,k}$. For the remaining states use

$$g_{E_B}^k = \begin{cases} (\lambda_k/\lambda_{B,*})h^{B,*}, & \lambda_{B,*} > 0, \\ 0, & \lambda_{B,*} = 0. \end{cases} \quad (19)$$

If the merged weight is zero, every constituent weight is zero as well. In every case $g_{E_B}^k \in \lambda_k K_B$, and the refined vectors sum to the block deviation. Put $g_e^k = 0$ on bridges and combine blocks. The block factorization gives $Ag^k = 0$. Then $f^k = \lambda_k v + g^k$ satisfies $Af^k = \lambda_k b$. Its capacity bounds hold on blocks by the scaled domains, and on bridges because $P \neq \emptyset$ implies $0 \leq v_e \leq u_e$. Equations (17) and (18), together with the bridge equations, give $\sum_k f^k = x$ and all observed products. Apply [proposition 2.1](#). $\square$

The domain condition (16) is itself linear:

$$A_B h^{B,k} = 0, \quad -\lambda_{B,k} v_{E_B} \leq h^{B,k} \leq \lambda_{B,k} (u_{E_B} - v_{E_B}). \quad (20)$$

Finite bounds force $h^{B,k} = 0$ at zero weight. No division by a variable is needed in the formulation; division appears only in the recovery proof at strictly positive weights. If $J_B = \emptyset$, the block has one merged state of weight one and its conditions reduce to $x_{E_B} - v_{E_B} \in K_B$, already enforced by $x \in P$. Thus unobserved blocks require no new variables or constraints.

The theorem is a constructive specialization of the known shared-simplex gluing principle. It does not introduce independent simplex vectors for different blocks: all use the same global y . It also does not merge observed state slices, whose domains can differ after the products are fixed.

Example 3.3 (A common matrix is not sufficient for merging). Let $M = (-1, 1, 2)^\top$, $c^1 = (0, 1, 100)^\top$, and $c^2 = (0, 100, 2)^\top$. Both polytopes $Q_i = \{q \in \mathbb{R} : Mq \leq c^i\}$ equal $[0, 1]$. For weights $\alpha_1 = \alpha_2 = 1/2$, the weighted Minkowski sum is still $[0, 1]$, whereas the aggregated rows $Mq \leq (c^1 + c^2)/2$ allow $q = 2$. The two upper bounds become $q \leq 101/2$ and $2q \leq 51$. Thus replacing state-specific systems by the sum of their right-hand sides is generally inexact, even for a common matrix and nonempty bounded domains. This is the standard reaggregation boundary discussed by Kis and Horváth [9, Section 2, equation (7)]. The sets in this example are identical, so the set identity (12) still holds. What fails is its replacement by addition of the different, redundant right-hand-side descriptions. In [theorem 3.2](#), every state instead uses the single fixed representation (10), with its bounds scaled by the state weight as in (20). Adding these scaled bounds gives that same representation scaled by the total weight, so the set identity justifies the merged formulation.

Finally, these results concern the hull of (3). Intersecting H with additional linear side constraints yields a valid relaxation of the graph with those constraints, but need not give its exact convex hull. Such constraints can couple blocks or alter conditional state domains; exactness then requires a separate argument.

4 Compression by cycles and observations

This section compresses the exact block formulation in two steps. First, cycle coordinates replace arc coordinates and suppress long paths. Second, observations determine some cycle coordinates, leaving only cycles supported on unobserved arcs. We assume $P \neq \emptyset$; if the original flow domain is empty, its hull is described by any inconsistent linear system.

4.1 Fixed coordinates and reduced graphs

The structural parameters can be applied after elementary exact preprocessing. This is useful when capacities force a nominal cycle to carry no freedom.

Lemma 4.1 (Deleting fixed arcs). *Let $F \subseteq E$ and suppose $x_e = c_e$ throughout P for every $e \in F$, where $c \in \mathbb{Q}^F$ is known. Set $R = E \setminus F$, $b' = b - A_{:,F}c$, and*

$$P' = \{x_R : A_{:,R}x_R = b', 0 \leq x_R \leq u_R\}, \quad \mathcal{O}' = \mathcal{O} \cap (R \times [m]).$$

The hull for $(P', \mathcal{O}')$, together with $x_F = c$ and $z_{ej} = c_e y_j$ for observed $e \in F$, is affinely isomorphic to H . If F contains every coordinate fixed throughout P , then

$$\text{aff } P' = \{x_R : A_{:,R}x_R = b'\}. \quad (21)$$

Proof. Feasible vectors of P restrict to P' . Conversely, adjoining c to a vector of P' satisfies the original balances and bounds, because $0 \leq c \leq u_F$. The same bijection at graph points adjoins the affine coordinates $z_{ej} = c_e y_j$, so it commutes with convexification.

For the last assertion, every remaining coordinate is nonconstant on P' . For each $e \in R$, choose two feasible vectors differing in coordinate e ; their midpoint has $0 < x_e < u_e$. Average these midpoints over $e \in R$. The resulting feasible vector is strictly inside every remaining bound. A sufficiently small relative neighborhood of that vector in the balance affine space therefore remains feasible, proving (21). If R is empty, both spaces are the same singleton and the assertion is immediate. $\square$

The lemma takes the fixed coordinates as input; it does not require a particular detection algorithm. Deleting them can split blocks or reduce their cycle ranks. Once all fixed coordinates are removed, the remaining cycle space is the direction space of the feasible affine hull. Conditional slices at a particular observation or zero state weight can still have smaller dimension.

4.2 Suppressing paths

Fix a cyclic block B . Degrees in this subsection are degrees *within the block*, with loops counted twice. For $r_B \geq 2$, suppress maximal paths whose internal vertices have degree two, and give each resulting core edge an arbitrary orientation. A rank-one block is a single cycle; represent its entire cyclic traversal by one core loop. This includes two parallel edges and an original self-loop.

Write $\mathcal{P}_B$ for the core edges, or equivalently the original paths, and $k_B = |\mathcal{P}_B|$. For an original arc e on path p , let $\varepsilon_e \in \{-1, 1\}$ compare its orientation with the chosen path traversal. A circulation has constant signed deviation on a path:

$$x_e = v_e + \varepsilon_e \delta_p \quad (e \in p). \quad (22)$$

The reason is balance of the block circulation at each internal degree-two vertex. Balances at the unsuppressed vertices say precisely that δ is a circulation of the core. Arc bounds give the path interval

$$\begin{aligned} L_p &= \max_{e \in p} \ell_e, & U_p &= \min_{e \in p} u'_e, \\ [\ell_e, u'_e] &= \begin{cases} [-v_e, u_e - v_e], & \varepsilon_e = 1, \\ [v_e - u_e, v_e], & \varepsilon_e = -1. \end{cases} \end{aligned} \quad (23)$$

All intervals are finite. Their joint feasibility follows from $P \neq \emptyset$.

Suppression preserves cycle rank. If $r_B \geq 2$, the connected core has minimum degree at least three: otherwise the original block would have been a single cycle. If its vertex count is n_B , then $2k_B \geq 3n_B$ and $r_B = k_B - n_B + 1$, so

$$n_B \leq 2r_B - 2, \quad k_B \leq 3r_B - 3. \quad (24)$$

For rank one, $k_B = 1$. In particular, $k_B \leq 3r_B$ in every case.

Choose a spanning tree of the core and let $\hat{C}_B \in \{0, \pm 1\}^{k_B \times r_B}$ be its fundamental-cycle matrix, normalized to have identity chord rows. Its columns parametrize core circulations bijectively. Thus the feasible core domain is

$$T_B = \{t \in \mathbb{R}^{r_B} : L_p \leq (\hat{C}_B t)_p \leq U_p \quad (p \in \mathcal{P}_B)\}. \quad (25)$$

It is nonempty and bounded, including in lower-dimensional cases. Boundedness follows already from the chord rows. For later use, choose one original representative arc $e(p)$ on each path and put

$$s_p(x) = \varepsilon_{e(p)}(x_{e(p)} - v_{e(p)}), \quad q_{ej} = \varepsilon_e(z_{ej} - v_e y_j). \quad (26)$$

For $x \in P$, $s_p(x)$ is its core deviation. A separate observation q_{ej} is retained for *every* observed original arc, including multiple observations of one path.

Theorem 4.2 (Cycle-coordinate extended hull). *The hull H has an exact rational extended formulation with*

$$\sum_{B \in \mathcal{B}} r_B a_B \quad (27)$$

additional variables and

$$O \left(|V| + |E| + m + |\mathcal{O}| + \sum_{B: a_B > 0} r_B (a_B + 1) \right) \quad (28)$$

rows, counting the original domain constraints. Coefficients of x , z , and additional variables can all be chosen from $\{0, \pm 1\}$. Coefficients of y and the right-hand sides may depend on the rational network data. All coefficient encoding lengths are polynomial in the input length.

Proof. Retain $x \in P$, $y \in \Delta_m$, and the bridge equations (15). For each B with $a_B > 0$, introduce $t^{B,j} \in \mathbb{R}^{r_B}$ for $j \in J_B$ and impose

$$y_j L_p \leq (\hat{C}_B t^{B,j})_p \leq y_j U_p, \quad j \in J_B, p \in \mathcal{P}_B, \quad (29)$$

$$q_{ej} = (\hat{C}_B t^{B,j})_p, \quad (e, j) \in \mathcal{O}, e \in p \subseteq E_B, \quad (30)$$

$$\lambda_{B,*} L_p \leq s_p(x) - \sum_{j \in J_B} (\hat{C}_B t^{B,j})_p \leq \lambda_{B,*} U_p, \quad p \in \mathcal{P}_B. \quad (31)$$

The aggregate deviation $s(x)$ is a core circulation. Hence the middle quantity in (31) is a core circulation as well. These constraints are exactly [theorem 3.2](#) in core coordinates, with the merged state eliminated by its sum equation. At a zero weight, the finite path bounds force every core path value to zero; the identity chord rows force the entire state vector to zero. Consequently the equivalence includes all simplex boundary strata. Blocks with $a_B = 0$ add no constraints, as in the block theorem.

There are $r_B a_B$ variables, $2k_B(a_B + 1)$ bound rows, and one equation per block observation. Bound (24) proves the row count. Every state variable has coefficient 0, 1, or -1 . An observation row has one original product coordinate, and each residual row uses one original flow coordinate through $s_p(x)$; no sums of different cycle expressions for that flow coordinate are necessary. This proves the stated coefficient bound. Reference flows are sums of rational balances; path endpoints are selected signed capacity offsets. The remaining coefficients are sums of these data, with polynomial rational encoding length. $\square$

4.3 Eliminating observed directions

Define the observed and unobserved arcs for a block and label by

$$O_{Bj} = \{e \in E_B : (e, j) \in \mathcal{O}\}, \quad U_{Bj} = E_B \setminus O_{Bj}.$$

Let $c(V_B, U_{Bj})$ count all connected components of this unobserved undirected subgraph, including isolated vertices. Its cycle rank is

$$\rho_{Bj} = |U_{Bj}| - |V_B| + c(V_B, U_{Bj}). \quad (32)$$

Loops and parallel edges keep their usual multigraph meanings. In particular, each unobserved loop contributes one to the cycle rank.

Lemma 4.3 (Unit minors for cycle elimination). *A fundamental-cycle matrix with identity chord rows is totally unimodular. Let C be such a matrix, let D consist of d linearly independent rows, and choose d columns I such that $B = D_{:,I}$ is nonsingular. Write F for the complementary columns. Then, for every row c of C , the vectors*

$$W = c_I B^{-1}, \quad R = c_F - W D_{:,F} \quad (33)$$

have entries in $\{0, \pm 1\}$.

Proof. Delete one incidence row of the connected core and partition the reduced incidence matrix as $[N_T \ N_Q]$ into tree and chord columns. A tree basis has determinant ± 1 . Every square incidence minor is 0 or ± 1 : induction expands a column with at most one nonzero entry; if every column has two entries, they are opposite and the row sum is zero. Every minor of $N_T^{-1} N_Q$ equals, up to sign, the determinant of a matrix obtained by replacing the corresponding tree columns by chord columns, divided by $\det N_T$. This follows by multiplying the replacement matrix by N_T^{-1} and expanding its unchanged identity columns. The replacement is a square incidence submatrix. Therefore $-N_T^{-1} N_Q$ is totally unimodular. Appending identity chord rows preserves this property, proving the assertion for C . A core consisting of one loop has $C = (1)$ directly.

In particular $\det B = \pm 1$. Cramer's rule applied to $WB = c_I$ expresses each entry of W as a selected-row replacement minor of C divided by $\det B$. For a free column f , the bordered determinant identity gives

$$R_f = \frac{\det \begin{pmatrix} B & D_{:,f} \\ c_I & c_f \end{pmatrix}}{\det B}.$$

The numerator is a minor of C , or zero when its rows repeat. These formulas prove (33). Empty pivot or free sets have their usual zero-dimensional matrix interpretation. $\square$

Theorem 4.4 (Observation-sensitive extended hull). *There is an exact rational extended formulation for H with*

$$\sum_{B \in \mathcal{B}} \sum_{j \in J_B} \rho_{Bj} \quad (34)$$

additional variables and the row bound (28). Coefficients of x , z , and additional variables belong to $\{0, \pm 1\}$; simplex coefficients and right-hand sides have polynomial rational encoding length. The variable count is an upper bound for this construction, not a lower bound for arbitrary extended formulations.

Proof. Fix B and $j \in J_B$, suppress their subscripts, and write $C = \widehat{C}_B$, $r = r_B$. Choose a maximal independent set of observed path rows of C , giving $D \in \mathbb{R}^{d \times r}$. For each selected row choose one actual observed arc and use its value from (26), giving $q \in \mathbb{R}^d$. The selected entries of q involve distinct original product variables. All other observations will remain as equations, including any repeated path values.

Choose pivot columns I and complementary columns F as in lemma 4.3. Retain $g = t_F \in \mathbb{R}^{r-d}$ and eliminate the pivot coordinates by the reversible substitution

$$t_I = B^{-1}(q - D_{:,F}g). \quad (35)$$

For every core path p with row c_p define

$$\phi_{pj} = (c_{p,I}B^{-1})q + (c_{p,F} - c_{p,I}B^{-1}D_{:,F})g. \quad (36)$$

Replace $(\widehat{C}_B t^{B,j})_p$ by ϕ_{pj} in (29)–(31). Selected observation equations become identities and can be omitted; all other observation equations remain. Formula (35) proves equivalence in both directions, so no projection condition is lost.

By lemma 4.3, each selected product and retained variable has coefficient 0, 1, or -1 in every ϕ_{pj} . A nonselected observation adds a distinct product coordinate; it cannot add a second occurrence of a selected product. Different labels use disjoint product and auxiliary columns. The residual uses one original flow coordinate through $s_p(x)$. Thus collecting terms does not increase any x , z , or auxiliary coefficient beyond one in magnitude. Only coefficients of y collect reference-flow and interval data. Their encoding lengths remain polynomial. Elimination adds no rows.

It remains to identify $r - d$. Restrict a block circulation to O_{Bj} . The kernel is precisely the space of circulations supported on U_{Bj} , whose dimension is (32) by the incidence-rank formula. The core matrix parametrizes all block circulations, and observing any arc of a suppressed path fixes its signed path value. Therefore the rank of the restriction map is exactly the rank d of the observed core rows. Rank-nullity gives $r_B - d = \rho_{Bj}$, proving (34). $\square$

For clarity, the final block inequalities are

$$y_j L_p \leq \phi_{pj} \leq y_j U_p, \quad j \in J_B, p \in \mathcal{P}_B, \quad (37)$$

$$q_{ej} = \phi_{pj}, \quad (e, j) \in \mathcal{O}, e \in p \subseteq E_B, \quad (38)$$

$$\lambda_{B,*} L_p \leq s_p(x) - \sum_{j \in J_B} \phi_{pj} \leq \lambda_{B,*} U_p, \quad p \in \mathcal{P}_B. \quad (39)$$

The original flow and simplex domains and bridge observations remain part of the formulation. In particular, reconstructing state products by balance equations alone is insufficient: all displayed state and residual bounds must also be imposed.

The row counts do not bound matrix nonzeros when ranks grow. If $N = |V| + |E| + m + |\mathcal{O}|$ and $\mathcal{O}_B = \mathcal{O} \cap (E_B \times [m])$, both constructions have the explicit nonzero bound

$$O \left(N + \sum_{B: a_B > 0} [r_B^2(a_B + 1) + r_B |\mathcal{O}_B|] \right). \quad (40)$$

Indeed, a reconstructed path has $O(r_B)$ selected-product and auxiliary terms per label; there are $O(r_B)$ paths and a_B explicit labels. Each remaining observation also uses $O(r_B)$ terms. Hence bounded maximum block rank gives size linear in the sparse input, even as m and the graph grow. The whole formulation need not be totally unimodular. The unit coefficient statements use the displayed rational row scaling; clearing denominators in the simplex coefficients can destroy that unit bound in a primitive integer normalization.

4.4 Forest complements and product completion

Corollary 4.5 (An original-coordinate hull on arbitrary graphs). *Suppose (V_B, U_{Bj}) is a forest for every B and $j \in J_B$. Then (37)–(39), the original domains, and the bridge equations describe H with no additional variables. They have unit flow and product coefficients and the row bound (28).*

Proof. An undirected multigraph is a forest exactly when its cycle rank is zero. Thus every ρ_{Bj} in (34) vanishes. Apply theorem 4.4. $\square$

Equivalently, the observations for each locally observed label meet every cycle of its block. Entirely unobserved labels impose no condition: they are already included in the merged state. The condition is sufficient; it does not assert that remaining unobserved cycles prevent an original-coordinate description with unit coefficients.

Proposition 4.6 (Minimum individual-coordinate completion). *Fix B and $j \in J_B$. Among additions of individual missing arc-product coordinates, exactly ρ_{Bj} are necessary and sufficient to determine the full state circulation from the observed coordinates and linear balance equations on the ambient circulation space. A minimum choice consists of the nonforest edges of any spanning forest of (V_B, U_{Bj}) . The additional variables in theorem 4.4 may instead be chosen to represent these individual missing products, with the same variable and row bounds and unit flow, product, and auxiliary coefficients.*

Proof. The existing restriction map has nullity ρ_{Bj} . Each additional scalar coordinate can raise its rank by at most one, proving necessity. Let F_{Bj} be a spanning forest of the unobserved subgraph, including all its vertices. Add observations on $U_{Bj} \setminus F_{Bj}$, which has ρ_{Bj} edges. The remaining unobserved graph is a forest, so the restriction kernel is zero by the proof of theorem 4.4. This proves sufficiency and the stated minimum.

Apply corollary 4.5 to the completed observation set and regard the added product coordinates as auxiliary variables. Projection onto the original observations commutes with convexification, so this is an extended formulation for the original H . Completing labels only in J_B does not introduce new labels. The added equations number $\sum_{B,j} \rho_{Bj} \leq \sum_B r_B a_B$, already absorbed by the row bound. The corollary's coefficient property includes the added coordinates. $\square$

There is a direct interpretation of this completion. The block state flows satisfy $A_B f^{B,j} = y_j A_B v_{E_B}$, since their deviations are block circulations. Once the nonforest state products are known, sum these equations on either side of a cut formed by deleting a remaining forest edge from its component. That edge is the only unknown crossing edge, so its state product is an affine expression

in the known state products and the state weight, with unit product coefficients. Component balance equations enforce consistency if the unobserved forest is disconnected. This is network elimination of the kind underlying reduced RLT: Liberti and Pantelides [10, Theorem 3.1 and Section 2] reconstruct common-factor products by multiplying full-row-rank linear equations and retaining products on suitable complementary coordinates. Here the extra information is the observation-complement graph criterion and its exact sparse simplex-hull consequence.

The minimum in [proposition 4.6](#) concerns reconstruction by *individual coordinates on the ambient linear space*. It is neither an extension-complexity lower bound nor a claim about the number of coordinates needed on a capacity-degenerate slice. In particular, $\lambda_j = 0$ forces every state product to zero. The preprocessing in [lemma 4.1](#) can remove fixed arcs before the parameters are computed, but it does not remove this conditional-slice distinction.

All formulations also give explicit recovery. From retained coordinates, recover each $t^{B,j}$ using (35), reconstruct path deviations by $\hat{C}_B t^{B,j}$, and subtract their sum from $s(x)$ to recover the merged state. Expand deviations along the original paths using their signs. The proportional refinement (19) then supplies the global state flows and the at-most- $m + 1$ -point decomposition from [proposition 2.1](#). This recovery is exact rational arithmetic for rational input and a rational feasible extension; merely solving the formulation numerically does not provide an exact certificate.

Example 4.7 (Three observations on a K_4 block). Orient every edge of K_4 from the smaller to the larger vertex and take unit capacities and unit flow from vertex 1 to vertex 4. Thus $b = (-1, 0, 0, 1)^\top$. Use the reference flow $v_{14} = 1$ and zero on the other arcs. For one label j , observe 13, 14, and 24; the unobserved edges 12, 23, and 34 form a spanning tree. Writing $q_{13} = z_{13,j}$, $q_{14} = z_{14,j} - y_j$, and $q_{24} = z_{24,j}$, balance gives the remaining state deviations

$$h_{12} = -q_{13} - q_{14}, \quad h_{23} = -q_{13} - q_{14} - q_{24}, \quad h_{34} = -q_{14} - q_{24}.$$

For example, nonnegativity of the reconstructed flow on 23 gives

$$z_{13,j} + z_{14,j} + z_{24,j} \leq y_j. \tag{41}$$

This cut is absent from independent product convexification. To see the gap, take $m = 1$, $y_1 = 1/2$, $z_{13,1} = z_{14,1} = z_{24,1} = 1/5$, and let x average the four paths 1234, 124, 134, and 14 equally. Then $x_{12} = x_{34} = 1/2$ and the other four arc flows are $1/4$. All three observed products satisfy (5), but the left side of (41) is $3/5 > 1/2$. The complete hull requires all reconstructed bounds, including the residual bounds, not only this one cut.

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